

Rogério Celestino

Senior Software Engineer · Transitioning into Applied AI (LLMs, RAG, On-Device ML) · MSc

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PROFESSIONAL PROFILE

Senior software engineer with 20+ years building and shipping production systems, now deliberately focused on Applied AI. I completed a postgraduate program in AI Engineering (2025) and moved into AI engineering through hands-on projects — LLM applications, retrieval-augmented search, and on-device inference — pairing them with two decades of production depth: the architecture, backend, and mobile delivery that turn models into real products. I'm targeting an Applied AI / AI Engineer role where engineering judgment matters as much as model knowledge. MSc in Software Engineering; former undergraduate and postgraduate lecturer; high-traffic fintech and marketplace background; five peer-reviewed publications.

CORE COMPETENCIES

Applied AI & LLMs: LLM application development, prompt & context engineering, structured/JSON outputs, function calling, retrieval-augmented generation (RAG), semantic search & embeddings (sentence-transformers, FAISS), on-device LLM inference (MediaPipe, Gemma), Gemini & OpenAI APIs, LangChain

Machine Learning & Data: Machine Learning, Deep Learning (PyTorch, TensorFlow, scikit-learn), Computer Vision (OpenCV), NLP, data pipelines & workflow orchestration (Prefect)

Backend & Cloud: Python, FastAPI, REST APIs, Firebase / Firestore, Google Cloud Functions, AWS, Docker, SQL, event-driven & serverless architectures

Mobile: iOS (Swift, SwiftUI, Objective-C), Android (Kotlin, Java, Jetpack Compose), Clean Architecture, MVVM / VIP, Server-Driven UI (SDUI), modularization, CI/CD (Bitrise, Fastlane)

Engineering & Leadership: Solution architecture, technical leadership, team building & mentoring, Agile / Scrum, TDD & software quality, R&D

AI PROJECTS (PERSONAL & PORTFOLIO)

- **Duo Concierge — AI Agent with Tool Use** (*Gemini · function calling · Streamlit · live demo*) — Conversational recommendation agent: an LLM with custom tools (geospatial Haversine filtering, schedule/availability checks) over a real dataset, deployed live on Streamlit. github.com/rogerioc/my-duogourmet-concierge-agent
- **AI Agents Engineering Guide — authored open guide** (*LangGraph · LlamaIndex · Graph RAG · evaluation & observability*) — Open technical guide on production agent engineering: ReAct & multi-agent orchestration, context engineering, evaluation (LLM-as-a-Judge, DeepEval), observability (Langfuse), and advanced / Graph / Agentic RAG (Neo4j, LlamaIndex). github.com/rogerioc/ai-agents-engineering-guide
- **AI Backend — RAG, LLM Orchestration & API** (*Python · FastAPI · LangChain · FAISS*) — Semantic search (transformer embeddings + FAISS), an LLM conversational assistant (LangChain + Gemini, memory, structured outputs), and a tested FastAPI service — with documented technology trade-offs. github.com/rogerioc/ai-backend-rag-llm
- **Vector-Ranking Reverse Engineering** (*Python · applied math*) — Reconstructed an opaque vector-database ranking function from oracle outputs, deriving its L_∞ normalization factor and penalty constant through systematic mathematical analysis. github.com/rogerioc/vector-ranking-study
- **Hybrid LLM Assistant — Fortuna (case study)** (*Gemini · Gemma on-device · MediaPipe · Cloud Functions*) — Hybrid cloud + on-device AI assistant: cloud inference with structured outputs plus on-device LLM (Gemma 3n via MediaPipe) in a Jetpack Compose Android client; event-driven backend orchestrated with Prefect. github.com/rogerioc/fortuna-ai-case-study

PROFESSIONAL EXPERIENCE

Software Engineer Specialist – Mobile — Inter

Jun 2025 – Jun 2026

- Led mobile architecture for a payment-terminal product, delivering a lean, purpose-built app through build flavors, selective dependency exclusion, and product variants with zero code duplication.
- Provided technical leadership on the Server-Driven UI (SDUI) SDK; worked across Kotlin/Swift, NFC, and SoftPOS (Inter Pag). R&D and continuous process improvement.

Mobile Engineer / Tech Lead — GetNinjas

2022 – 2025

- Led squad technical direction, solution architecture, and strategic mobile planning; drove reliability and scaling (achieved high crash-free rates) integrating AWS, Firebase, Salesforce CMS and attribution platforms (Singular, AppsFlyer).
- Delivered iOS features in Swift with Clean Architecture, VIP, modularization, and CI/CD (Bitrise); mentored the team and optimized architecture.

Mobile Specialist — Zup Innovation

2021 – 2022

- Architected Android and iOS solutions, delivering clean, well-designed code in small functional increments and driving key implementation decisions.

iOS Specialist — Warren Brasil

2020 – 2021

- Built and maintained features in Swift / SwiftUI applying modern architectures (MVVM, The Composable Architecture) with a strong unit-testing focus.

Mobile Solutions Architect / iOS Tech Lead — Inter

2019 – 2020

- Led the iOS team and architected Android, iOS, and back-end solutions; owned the Inter Empresas app and contributed to PIX, Brazil's instant-payment system. Built a WebView micro-framework to accelerate feature delivery across native platforms.

Senior Mobile Software Engineer — MaxMilhas

2018 – 2019

- Developed native Android (Kotlin/Java) and iOS (Swift/Objective-C) apps with reactive programming (Rx); served as technical reference and stepped in as Tech Lead within six months. App drove 20%+ of ticket sales.

Earlier roles (2003–2018): Mobile Tech Lead @ Samba Tech (built the mobile team from scratch) · R&D Researcher @ Netbee (CNPq, Intelligent Transport Systems, MQTT) · University Professor (PUC Minas, CEFET-MG, UniBH — mobile architecture, data structures, 3D programming) · Engineer @ Stefanini, TACOM, Pontomobi, Avenue Code · QA @ CIn/Motorola-UFPE. Full history on LinkedIn.

EDUCATION

AI Data Engineer 4.0 — professional program — Data Science Academy

2026 – in progress

LLM data pipelines, embeddings & vector DBs, RAG / GraphRAG / Agentic RAG, ETL for unstructured data (516h).

Postgraduate Program in AI Engineering (Lato Sensu) — Data Science Academy

2025 – 2025

MEC-recognized specialist degree — Machine Learning, Deep Learning & applied AI engineering (384h).

MSc, Software Engineering — PUC Minas

2007 – 2009

Mobile software engineering, software product lines, feature- and aspect-oriented programming.

Specialization, Software Test Engineering — CIn / UFPE (Motorola partnership)

2006

BSc, Information Systems — PUC Minas

2002 – 2005

SELECTED CERTIFICATIONS

- Postgraduate module in AI Engineering, Generative AI & LLMs for NLP, Deep Learning with Python & C++ (PyTorch / TensorFlow), AI for Computer Vision, Financial Engineering with AI — Data Science Academy
- AWS Generative AI Bootcamp — Insper · Machine Learning A-Z (AI, Python & R) — Udemy · Building a Project Portfolio with GitHub — Data Science Academy

PUBLICATIONS & LANGUAGES

Publications: Five peer-reviewed papers (2005–2008) on software product lines, feature- and aspect-oriented programming, and data-structure optimization for mobile/embedded systems — SBGames, LA-WASP, UFPE, EMECOMP, SIRC/RS.

Languages: Portuguese (native) · English (professional working proficiency).